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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Magnetic Properties of Matter

JEE Main Physics · Previous Year Questions with Answer Key

84	73	2003 – 2026	18	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer) 54 questions

Q1. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . The magnetic intensity in SI units is $\alpha \times 10^5$. The value of α is ____ .

(Free space permeability $\mu_0 = 4\pi \times 10^{-7}$ SI units.)

(A) $\frac{25}{\pi}$ (B) $\frac{1}{16\pi}$ (C) $\frac{1}{\pi}$ (D) $\frac{1}{4\pi}$

Q2. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Consider a ferromagnetic material :

Assertion (A) : The individual atoms in a ferromagnetic material possess a magnetic dipole moment and interact with one another in such a way that they spontaneously align themselves forming domains.

Reason (R) : At high enough temperature, the domain structure of ferromagnetic material disintegrates. Thus, magnetization will disappear at high enough temperature known as Curie temperature.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

Q3. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A): Magnetic monopoles do not exist.

Reason (R): Magnetic field lines are continuous and form closed loops.

In the light of the above statements, choose the **most appropriate answer** from the options given below:

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) (A) is not correct but (R) is correct

(C) (A) is correct but (R) is not correct

(D) Both (A) and (R) are correct but (R) is **not** the correct explanation of (A)

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Q4. JEE Main 2025 (Online) 7th April Morning Shift

The percentage increase in magnetic field (B) when space within a current carrying solenoid is filled with magnesium (magnetic susceptibility $\chi_{Mg} = 1.2 \times 10^{-5}$) is :

(A) $\frac{5}{3} \times 10^{-5}\%$ (B) $\frac{5}{6} \times 10^{-4}\%$ (C) $\frac{5}{6} \times 10^{-5}\%$ (D) $\frac{6}{5} \times 10^{-3}\%$

Open solution

Q5. JEE Main 2025 (Online) 3rd April Evening Shift

A magnetic dipole experiences a torque of $80\sqrt{3}\,\mathrm{N\,m}$ when placed in uniform magnetic field in such a way that dipole moment makes angle of 60° with magnetic field. The potential energy of the dipole is :

(A) $-40\sqrt{3}\,\mathrm{J}$ (B) -80 J

(C) 80 J (D) -60 J

Open solution

Q6. JEE Main 2025 (Online) 2nd April Morning Shift

The relationship between the magnetic susceptibility (χ) and the magnetic permeability (μ) is given by :

(μ_0 is the permeability of free space and μ_r is relative permeability)

(A) $\chi = \frac{\mu}{\mu_0} - 1$ (B) $\chi = \frac{\mu_r}{\mu_0} + 1$

(C) $\chi = \mu_r + 1$ (D) $\chi = 1 - \frac{\mu}{\mu_0}$

Open solution

Q7. JEE Main 2025 (Online) 29th January Evening Shift

Match List - I with List - II.

List - I	List - II
(A) Magnetic induction	(I) Ampere meter^2
(B) Magnetic intensity	(II) Weber
(C) Magnetic flux	(III) Gauss
(D) Magnetic moment	(IV) Ampere meter^{-1}

Choose the correct answer from the options given below:

(A) (A)-(III), (B)-(IV), (C)-(II), (D)-(I) (B) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II) (D) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Open solution

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Q8. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Paramagnetic substances:

A. align themselves along the directions of external magnetic field.

B. attract strongly towards external magnetic field.

C. has susceptibility little more than zero.

D. move from a region of strong magnetic field to weak magnetic field.

Choose the most appropriate answer from the options given below:

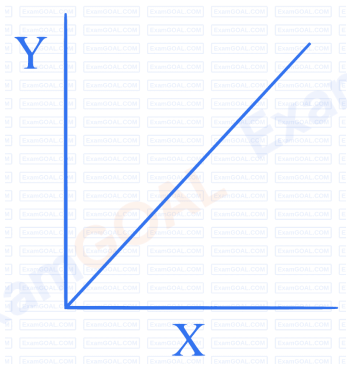
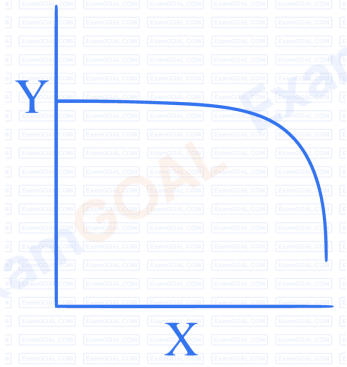
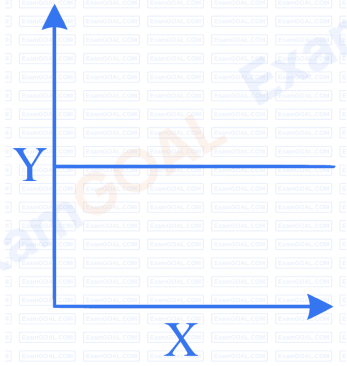
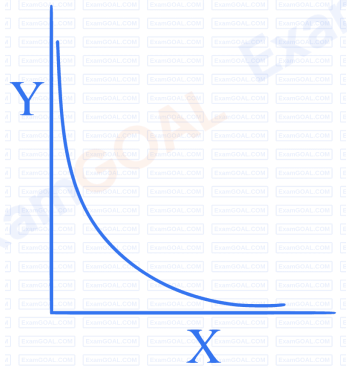
(A) A, B, C Only **(B)** A, C Only **(C)** A, B, C, D **(D)** B, D Only

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Q9. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

Match List I with List II:

	LIST I (Y vs X)		LIST II (Shape of Graph)
A.	Y = magnetic susceptibility X = magnetising field	I.	
B.	Y = magnetic field X = distance from centre of a current carrying wire for x < a (where a = radius of wire)	II.	
C.	Y = magnetic field $\mathrm{X}=$ distance from centre of a current carrying wire for $x>\mathrm{a}$ (where $\mathrm{a}=$ radius of wire)	III.	
D.	Y = magnetic field inside solenoid X = distance from centre	IV.	

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Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(IV), (C)- (I), (D)-(II)

(B) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(C) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(D) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

Q10. JEE Main 2024 (Online) 4th April Evening Shift

The magnetic moment of a bar magnet is $0.5\ \mathrm{Am}^2$. It is suspended in a uniform magnetic field of $8 \times 10^{-2}\ \mathrm{T}$. The work done in rotating it from its most stable to most unstable position is:

(A) $4 \times 10^{-2}\ \mathrm{J}$ (B) $16 \times 10^{-2}\ \mathrm{J}$ (C) $8 \times 10^{-2}\ \mathrm{J}$ (D) Zero

Open solution

Q11. JEE Main 2023 (Online) 12th April Morning Shift

Given below are two statements:

Statement I : The diamagnetic property depends on temperature.

Statement II : The induced magnetic dipole moment in a diamagnetic sample is always opposite to the magnetizing field.

In the light of given statements, choose the correct answer from the options given below.

(A) Statement I is incorrect but Statement II is true.

(B) Statement I is correct but Statement II is false.

(C) Both Statement I and Statement II are False.

(D) Both Statement I and Statement II are true.

Open solution

Q12. JEE Main 2023 (Online) 11th April Morning Shift

The free space inside a current carrying toroid is filled with a material of susceptibility 2×10^{-2}. The percentage increase in the value of magnetic field inside the toroid will be

(A) 0.1% (B) 1% (C) 2% (D) 0.2%

Open solution

Q13. JEE Main 2023 (Online) 10th April Evening Shift

Given below are two statements:

Statement I : For diamagnetic substance, $-1 \leq \chi < 0$, where χ is the magnetic susceptibility.

Statement II : Diamagnetic substances when placed in an external magnetic field, tend to move from stronger to weaker part of the field.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Statement I is correct but Statement II is false

(C) Both Statement I and Statement II are False

(D) Statement I is incorrect but Statement II is true.

Open solution

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Q14. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : Electromagnets are made of soft iron.

Reason R : Soft iron has high permeability and low retentivity.

In the light of above, statements, choose the most appropriate answer from the options given below.

(A) Both A and R are correct and R is the correct explanation of A

(B) A is not correct but R is correct

(C) A is correct but R is not correct

(D) Both A and R are correct but R is NOT the correct explanation of A

Q15. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

A bar magnet with a magnetic moment $5.0\mathrm{Am}^2$ is placed in parallel position relative to a magnetic field of $0.4\ \mathrm{T}$. The amount of required work done in turning the magnet from parallel to antiparallel position relative to the field direction is _____.

(A) zero (B) 1 J (C) 2 J (D) 4 J

Q16. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

A rod with circular cross-section area $2\ \mathrm{cm}^2$ and length $40\ \mathrm{cm}$ is wound uniformly with 400 turns of an insulated wire. If a current of $0.4\ \mathrm{A}$ flows in the wire windings, the total magnetic flux produced inside windings is $4\pi \times 10^{-6}\ \mathrm{Wb}$. The relative permeability of the rod is

(Given : Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}\mathrm{NA}^{-2}$)

(A) $\frac{5}{16}$ (B) 125 (C) $\frac{32}{5}$ (D) 12.5

Q17. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

A solenoid of 1200 turns is wound uniformly in a single layer on a glass tube 2 m long and 0.2 m in diameter. The magnetic intensity at the center of the solenoid when a current of 2 A flows through it is :

(A) $1\ \mathrm{A\ m^{-1}}$ (B) $2.4 \times 10^{-3}\ \mathrm{A\ m^{-1}}$

(C) $1.2 \times 10^{3}\ \mathrm{A\ m^{-1}}$ (D) $2.4 \times 10^{3}\ \mathrm{A\ m^{-1}}$

Q18. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

The space inside a straight current carrying solenoid is filled with a magnetic material having magnetic susceptibility equal to 1.2×10^{-5}. What is fractional increase in the magnetic field inside solenoid with respect to air as medium inside the solenoid?

(A) 1.2×10^{-5} (B) 1.2×10^{-3} (C) 1.8×10^{-3} (D) 2.4×10^{-5}

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Q19. JEE Main 2022 (Online) 27th June Morning Shift

The susceptibility of a paramagnetic material is 99. The permeability of the material in Wb/A-m, is :

[Permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ Wb/A-m]

(A) $4\pi \times 10^{-7}$ (B) $4\pi \times 10^{-4}$ (C) $4\pi \times 10^{-5}$ (D) $4\pi \times 10^{-6}$

Open solution

Q20. JEE Main 2022 (Online) 26th June Evening Shift

A bar magnet having a magnetic moment of 2.0×10^5 JT^{-1}, is placed along the direction of uniform magnetic field of magnitude B = 14×10^{-5} T. The work done in rotating the magnet slowly through 60° from the direction of field is :

(A) 14 J (B) 8.4 J (C) 4 J (D) 1.4 J

Open solution

Q21. JEE Main 2022 (Online) 25th June Evening Shift

Given below are two statements :

Statement I : Susceptibilities of paramagnetic and ferromagnetic substances increase with decrease in temperature.

Statement II : Diamagnetism is a result of orbital motions of electrons developing magnetic moments opposite to the applied magnetic field.

Choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Open solution

Q22. JEE Main 2022 (Online) 24th June Evening Shift

The soft-iron is a suitable material for making an electromagnet. This is because soft-iron has

(A) low coercivity and high retentivity. (B) low coercivity and low permeability.

(C) high permeability and low retentivity. (D) high permeability and high retentivity.

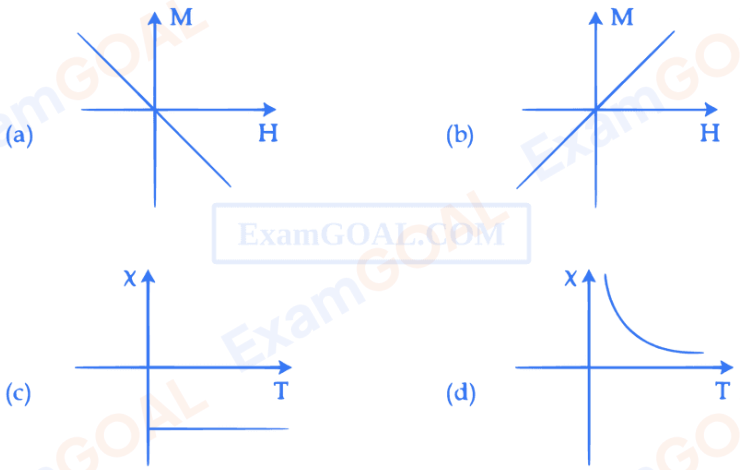
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Q23. JEE Main 2021 (Online) 1st September Evening Shift

Open solution

Following plots show magnetization (M) vs magnetizing field (H) and magnetic susceptibility (χ) vs temperature (T) graph :



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Which of the following combination will be represented by a diamagnetic material?

(A) (a), (c) (B) (a), (d) (C) (b), (d) (D) (b), (c)

Q24. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

Statement I : The ferromagnetic property depends on temperature. At high temperature, ferromagnet becomes paramagnet.

Statement II : At high temperature, the domain wall area of a ferromagnetic substance increases.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q25. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

The magnetic susceptibility of a material of a rod is 499. Permeability in vacuum is $4\pi \times 10^{-7}$ H/m. Absolute permeability of the material of the rod is :

(A) $4\pi \times 10^{-4}$ H/m (B) $2\pi \times 10^{-4}$ H/m (C) $3\pi \times 10^{-4}$ H/m (D) $\pi \times 10^{-4}$ H/m

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Q26. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

Which of the following statements are correct?

(A) Electric monopoles do not exist whereas magnetic monopoles exist.

(B) Magnetic field lines due to a solenoid at its ends and outside cannot be completely straight and confined.

(C) Magnetic field lines are completely confined within a toroid.

(D) Magnetic field lines inside a bar magnet are not parallel.

(E) $\chi = -1$ is the condition for a perfect diamagnetic material, where χ is its magnetic susceptibility.

Choose the correct answer from the options given below :

(A) (A) and (B) only **(B)** (B) and (C) only **(C)** (C) and (E) only **(D)** (B) and (D) only

Q27. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

In a ferromagnetic material, below the curie temperature, a domain is defined as :

(A) a macroscopic region with zero magnetization.

(B) a macroscopic region with saturation magnetization.

(C) a macroscopic region with randomly oriented magnetic dipoles.

(D) a macroscopic region with consecutive magnetic dipoles oriented in opposite direction.

Q28. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

A soft ferromagnetic material is placed in an external magnetic field. The magnetic domains :

(A) may increase or decrease in size and change its orientation.

(B) increase in size but no change in orientation.

(C) decrease in size and changes orientation.

(D) have no relation with external magnetic field.

Q29. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

An iron rod of volume 10^{-3} m^3 and relative permeability 1000 is placed as core in a solenoid with 10 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment of the rod will be :

(A) 5×10^2 Am^2 **(B)** 0.5×10^2 Am^2 **(C)** 500×10^2 Am^2 **(D)** 50×10^2 Am^2

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Q30. JEE Main 2020 (Online) 4th September Evening Slot

A paramagnetic sample shows a net magnetisation of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetisation will be:

Open solution

(A) 4 A/m (B) 1 A/m (C) 0.75 A/m (D) 2.25 A/m

Q31. JEE Main 2020 (Online) 4th September Morning Slot

A small bar magnet placed with its axis at 30° with an external field of 0.06 T experiences a torque of 0.018 Nm. The minimum work required to rotate it from its stable to unstable equilibrium position is :

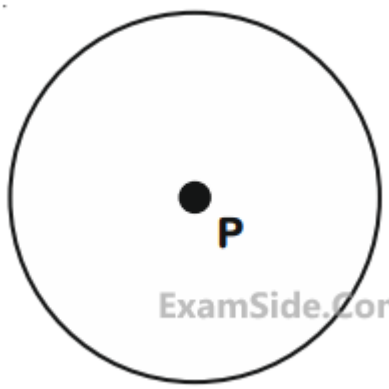
Open solution

(A) 6.4×10^{-2} J (B) 9.2×10^{-3} J (C) 7.2×10^{-2} J (D) 11.7×10^{-3} J

Q32. JEE Main 2020 (Online) 3rd September Evening Slot

A perfectly diamagnetic sphere has a small spherical cavity at its centre, which is filled with a paramagnetic substance. The whole system is placed in a uniform magnetic field $\vec{B}$. Then the field inside the paramagnetic substance is :

Open solution



(A) $\vec{B}$

(B) zero

(C) much large than $\left|\vec{B}\right|$ and parallel to $\vec{B}$

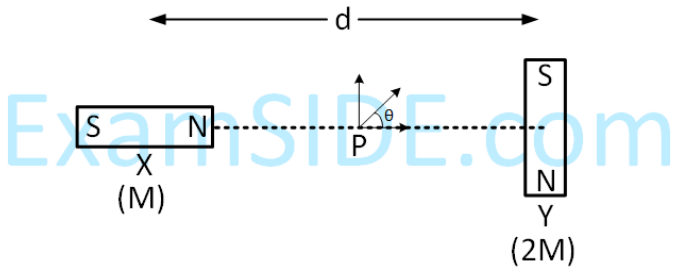
(D) much large than $\left|\vec{B}\right|$ but opposite to $\vec{B}$

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Q33. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

Two magnetic dipoles X and Y are placed at a separation d, with their axes perpendicular to each other. The dipole moment of Y is twice that of X. A particle of charge q is passing, through their midpoint P, at angle $\theta = 45°$ with the horizontal line, as shown in figure. What would be the magnitude of force on the particle at that instant ?
(d is much larger than the dimensions of the dipole)



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(A) $\left(\frac{\mu_0}{4\pi}\right)\frac{2M}{(d/2)^3} \times qv$

(B) $\left(\frac{\mu_0}{4\pi}\right)\frac{M}{(d/2)^3} \times qv$

(C) $\sqrt{2}\left(\frac{\mu_0}{4\pi}\right)\frac{M}{(d/2)^3} \times qv$

(D) 0

Q34. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

A paramagnetic material has 10^{28} atoms/m^3. Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4}. Its susceptibility at 300 K is :

(A) 3.726×10^{-4} (B) 2.672×10^{-4} (C) 3.672×10^{-4} (D) 3.267×10^{-4}

Q35. JEE Main 2019 (Online) 11th January Evening Slot

Open solution

A paramagnetic substance in the form of a cube with sides 1 cm has a magnetic dipole moment of 20×10^{-6} J/ T when a magnetic intensity of 60×10^3 A/m is applied. Its magnetic susceptibility is

(A) 3.3×10^{-4} (B) 2.3×10^{-2} (C) 4.3×10^{-2} (D) 3.3×10^{-2}

Q36. JEE Main 2019 (Online) 10th January Morning Slot

Open solution

A magnet of total magnetic moment $10^{-2}\,\hat{i}$ A-m^2 is placed in a time varying magnetic field, $B\hat{i}$ (cos ωt) where B = 1 Tesla and ω = 0.125 rad/s. The work done for reversing the direction of the magnetic moment at t = 1 second, is -

(A) 0.014 J (B) 0.028 J (C) 0.01 J (D) 0.007 J

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Q37. JEE Main 2019 (Online) 9th January Morning Slot

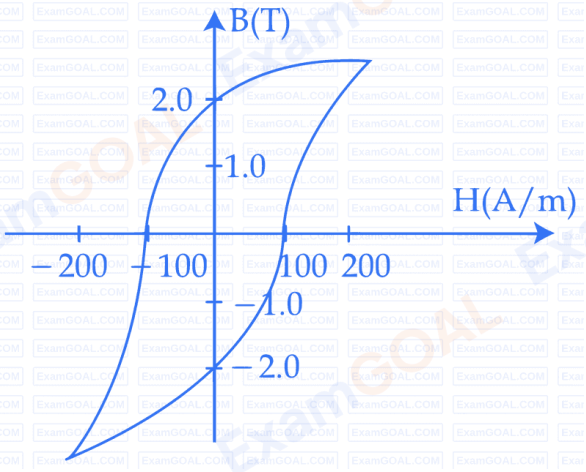
A bar magnet is demagnetized by inserting it inside a solenoid of length 0.2 m, 100 turns, and carrying a current of 5.2 A. The coercivity of the bar magnet is :

(A) 285 A/m (B) 2600 A/m (C) 520 A/m (D) 1200 A/m

Open solution

Q38. JEE Main 2018 (Online) 15th April Morning Slot

The B-H curve for a ferromagnet is shown in the figure. The ferromagnet is placed inside a long solenoid with 1000 turns/cm. The current that should be passed in the solenoid to demagnetise the ferromagnet completely is :



(A) $1\ mA$ (B) $2\ mA$ (C) $20\ \mu A$ (D) $40\ \mu A$

Open solution

Q39. JEE Main 2017 (Offline)

A magnetic needle of magnetic moment 6.7×10^{-2} A m^2 and moment of inertia 7.5×10^{-6} kg m^2 is performing simple harmonic oscillations in a magnetic field of 0.01 T. Time taken for 10 complete oscillations is:

(A) 8.76 s (B) 6.65 s (C) 8.89 s (D) 6.98 s

Open solution

Q40. JEE Main 2015 (Online) 11th April Morning Slot

A short bar magnet is placed in the magnetic meridian of the earth with north pole pointing north. Neutral points are found at a distance of 30 cm from the magnet on the east - West line, drawn through the middle point of the magnet. The magnetic moment of the magnet in Am^2 is close to :

(Given $\frac{\mu_0}{4\pi} = 10^{-7}$ in SI units and B_H = Horizontal component of earth's magnetic field = 3.6×10^{-5} Tesla.)

(A) 9.7 (B) 4.9 (C) 19.4 (D) 14.6

Open solution

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Q41. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

An example of a perfect diamagnet is a superconductor. This implies that when a superconductor is put in a magnetic field of intensity B, the magnetic field B_s inside the superconductor will be such that :

(A) $B_s = -B$

(B) $B_s = 0$

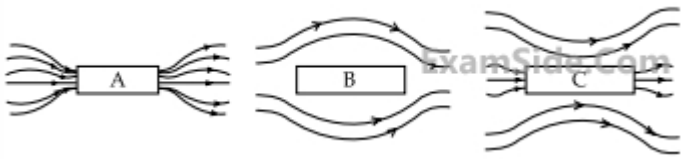
(C) $B_s = B$

(D) $B_s < B$ but $B_s \neq 0$

Q42. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

Three identical bars A, B and C are made of different magnetic materials. When kept in a uniform magnetic field, the field lines around them look as follows :



Make the correspondence of these bars with their material being diamagnetic (D), ferromagnetic (F) and paramagnetic (P) :

(A) A $\leftrightarrow$ D, B $\leftrightarrow$ P, C $\leftrightarrow$ F

(B) A $\leftrightarrow$ F, B $\leftrightarrow$ D, C $\leftrightarrow$ P

(C) A $\leftrightarrow$ P, B $\leftrightarrow$ F, C $\leftrightarrow$ D

(D) A $\leftrightarrow$ F, B $\leftrightarrow$ P, C $\leftrightarrow$ D

Q43. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

The mid points of two small magnetic dipoles of length d in end-on positions, are separated by a distance x, (x > > d). The force between them is proportional to x^{-n} where n is :



(A) 1

(B) 2

(C) 3

(D) 4

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Q44. JEE Main 2014 (Offline)

The coercivity of a small magnet where the ferromagnet gets demagnetized is $3 \times 10^3\ Am^{-1}$. The current required to be passed in a solenoid of length $10\ cm$ and number of turns 100, so that the magnet gets demagnetized when inside the solenoid, is :

Open solution

(A) $30\ mA$ (B) $60\ mA$ (C) $3\ A$ (D) $6A$

Q45. JEE Main 2013 (Offline)

Two short bar magnets of length $1\ cm$ each have magnetic moments $1.20\ Am^2$ and $1.00\ Am^2$ respectively. They are placed on a horizontal table parallel to each other with their N poles pointing towards the South. They have a common magnetic equator and are separated by a distance of $20.0\ cm$. The value of the resultant horizontal magnetic induction at the mid-point O of the line joining their centres is close to (Horizontal component of earth's magnetic induction is $3.6 \times 10^{-5} Wb/m^2$)

Open solution

(A) $3.6 \times 10^{-5}\ Wb/m^2$ (B) $2.56 \times 10^{-4}\ Wb/m^2$
(C) $3.50 \times 10^{-4}\ Wb/m^2$ (D) $5.80 \times 10^{-4}\ Wb/m^2$

Q46. AIEEE 2012 (Online) 7th May

A bar magnet of length $6\ \mathrm{cm}$ has a magnetic moment of $4\ \mathrm{J\ T^{-1}}$. Find the strength of magnetic field at a distance of $200\ \mathrm{cm}$ from the centre of the magnet along its equatorial line.

Open solution

(A) 4×10^{-8} tesla (B) 3.5×10^{-8} tesla (C) 5×10^{-8} tesla (D) 3×10^{-8} tesla

Q47. AIEEE 2008

Relative permittivity and permeability of a material ε_r and μ_r, respectively. Which of the following values of these quantities are allowed for a diamagnetic material?

Open solution

(A) $\varepsilon_r = 0.5,\ \mu_r = 1.5$ (B) $\varepsilon_r = 1.5,\ \mu_r = 0.5$
(C) $\varepsilon_r = 0.5,\ \mu_r = 0.5$ (D) $\varepsilon_r = 1.5,\ \mu_r = 1.5$

Q48. AIEEE 2006

Needles N_1, N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will

Open solution

(A) attract N_1 and N_2 strongly but repel N_3
(B) attract N_1 strongly, N_2 weakly and repel N_3 weakly
(C) attract N_1 strongly, but repel N_2 and N_3 weakly
(D) attract all three of them

Q49. AIEEE 2005

A magnetic needle is kept in a non-uniform magnetic field. It experiences :

Open solution

(A) neither a force nor a torque (B) a torque but not a force
(C) a force but not a torque (D) a force and a torque

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Q50. AIEEE 2004

Curie temperature is the temperature above which

(A) a ferromagnetic material becomes paramagnetic

(B) a paramagnetic material becomes diamagnetic

(C) a ferromagnetic material becomes diamagnetic

(D) a paramagnetic material becomes ferromagnetic

Open solution

Q51. AIEEE 2004

The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is $2s$. The magnet is cut along its length into three equal parts and these parts are then placed on each other with their like poles together. The time period of this combination will be

(A) $2\sqrt{3}\,s$ (B) $\frac{2}{3}\,s$ (C) $2\,s$ (D) $\frac{2}{\sqrt{3}}\,s$

Open solution

Q52. AIEEE 2004

The materials suitable for making electromagnets should have

(A) high retentivity and low coercivity (B) low retentivity and low coercivity

(C) high retentivity and high coercivity (D) low retentivity and high coercivity

Open solution

Q53. AIEEE 2003

A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60°. The torque needed to maintain the needle in this position will be :

(A) $\sqrt{3}\,W$ (B) W (C) $\frac{\sqrt{3}}{2}W$ (D) $2W$

Open solution

Q54. AIEEE 2003

The magnetic lines of force inside a bar magnet

(A) are from north-pole to south-pole of the magnet

(B) do not exist

(C) depend upon the area of cross-section of the bar magnet

(D) are from south-pole to north-pole of the Magnet

Open solution

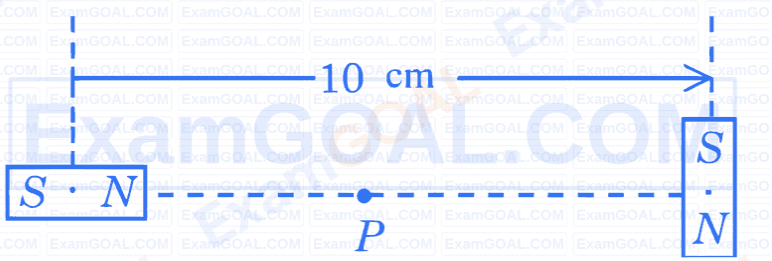
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Numerical 11 questions

Q55. JEE Main 2026 (Online) 4th April Evening Shift

Two identical small bar magnets each of dipole moment $3\sqrt{5}\,\mathrm{J/T}$ are placed at a center to center separation of 10 cm , with their axes perpendicular to each other as shown in figure. The value of magnetic field at the point P midway between the magnets is $\alpha \times 10^{-3}\,\mathrm{T}$. The value of α is ____

$(\mu_0 = 4\pi \times 10^{-7}\mathrm{Tm/A})$



Open solution

Q56. JEE Main 2026 (Online) 24th January Morning Shift

A short bar magnet placed with its axis at 30° with an external field of 800 Gauss, experiences a torque of $0.016\,\mathrm{N.m}$. The work done in moving it from most stable to most unstable position is $\alpha \times 10^{-3}\,\mathrm{J}$. The value of α is ____ .

Open solution

Q57. JEE Main 2024 (Online) 9th April Evening Shift

A straight magnetic strip has a magnetic moment of $44\,\mathrm{Am}^2$. If the strip is bent in a semicircular shape, its magnetic moment will be _____ Am^2.

(given $\pi = \frac{22}{7}$)

Open solution

Q58. JEE Main 2024 (Online) 8th April Evening Shift

The coercivity of a magnet is $5 \times 10^3\,\mathrm{A/m}$. The amount of current required to be passed in a solenoid of length $30\,\mathrm{cm}$ and the number of turns 150, so that the magnet gets demagnetised when inside the solenoid is _____ A.

Open solution

Q59. JEE Main 2024 (Online) 29th January Morning Shift

The magnetic potential due to a magnetic dipole at a point on its axis situated at a distance of $20\,\mathrm{cm}$ from its center is $1.5 \times 10^{-5}\,\mathrm{T\,m}$. The magnetic moment of the dipole is _____ $A\,\mathrm{m}^2$.
(Given : $\frac{\mu_o}{4\pi} = 10^{-7}\mathrm{Tm}A^{-1}$)

Open solution

Q60. JEE Main 2023 (Online) 10th April Morning Shift

The current required to be passed through a solenoid of 15 cm length and 60 turns in order of demagnetise a bar magnet of magnetic intensity $2.4 \times 10^3\ Am^{-1}$ is _____ A.

Open solution

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Q61. JEE Main 2023 (Online) 8th April Morning Shift

The magnetic intensity at the center of a long current carrying solenoid is found to be $1.6 \times 10^3 \mathrm{Am}^{-1}$. If the number of turns is 8 per cm, then the current flowing through the solenoid is _____ A.

Open solution

Q62. JEE Main 2021 (Online) 31st August Evening Shift

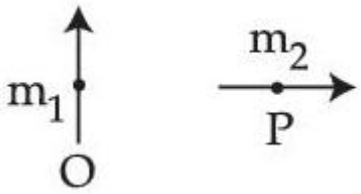
A long solenoid with 1000 turns/m has a core material with relative permeability 500 and volume 10^3 cm^3. If the core material is replaced by another material having relative permeability of 750 with same volume maintaining same current of 0.75 A in the solenoid, the fractional change in the magnetic moment of the core would be approximately $\left(\frac{x}{499}\right)$. Find the value of x.

Open solution

Q63. JEE Main 2021 (Online) 26th August Morning Shift

Two short magnetic dipoles m_1 and m_2 each having magnetic moment of 1 Am^2 are placed at point O and P respectively. The distance between OP is 1 meter. The torque experienced by the magnetic dipole m_2 due to the presence of m_1 is ..... $\times 10^{-7}$ Nm.

Open solution



Q64. JEE Main 2021 (Online) 27th July Morning Shift

In a uniform magnetic field, the magnetic needle has a magnetic moment 9.85 $\times 10^{-2}$ A m^2 and moment of inertia 5 $\times 10^{-6}$ kg m^2. If it performs 10 complete oscillations in 5 seconds then the magnitude of the magnetic field is _____ mT. [Take π^2 as 9.85]

Open solution

Q65. JEE Main 2021 (Online) 25th July Morning Shift

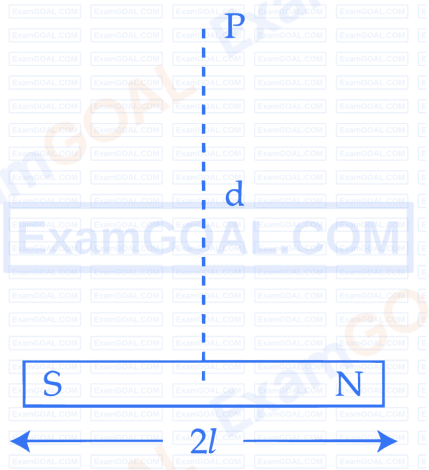
The value of aluminium susceptibility is 2.2 $\times 10^{-5}$. The percentage increase in the magnetic field if space within a current carrying toroid is filled with aluminium is $\frac{x}{10^4}$. Then the value of x is _____.

Open solution

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MCQ (More than One Correct Answer) 1 questions

Q66. JEE Main 2025 (Online) 28th January Evening Shift



Open solution

A bar magnet has total length $2l = 20$ units and the field point P is at a distance $\mathrm{d} = 10$ units from the centre of the magnet. If the relative uncertainty of length measurement is 1%, then uncertainty of the magnetic field at point P is :

(A) 10% (B) 5% (C) 3% (D) 4%

Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q67. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

The vertical component of the earth's magnetic field is $6 \times 10^{-5}\ \mathrm{T}$ at any place where the angle of dip is 37°. The earth's resultant magnetic field at that place will be (Given $\tan 37^{\circ} = \frac{3}{4}$)

(A) $8 \times 10^{-5}\ \mathrm{T}$ (B) $6 \times 10^{-5}\ \mathrm{T}$ (C) $5 \times 10^{-4}\ \mathrm{T}$ (D) $1 \times 10^{-4}\ \mathrm{T}$

Open solution

Q68. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

A compass needle of oscillation magnetometer oscillates 20 times per minute at a place P of dip 30°. The number of oscillations per minute become 10 at another place Q of 60° dip. The ratio of the total magnetic field at the two places $(B_Q : B_P)$ is :

(A) $\sqrt{3} : 4$ (B) $4 : \sqrt{3}$ (C) $\sqrt{3} : 2$ (D) $2 : \sqrt{3}$

Open solution

Q69. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Two bar magnets oscillate in a horizontal plane in earth's magnetic field with time periods of $3\ \mathrm{s}$ and $4\ \mathrm{s}$ respectively. If their moments of inertia are in the ratio of $3 : 2$, then the ratio of their magnetic moments will be:

(A) 2 : 1 (B) 8 : 3 (C) 1 : 3 (D) 27 : 16

Open solution

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Q70. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

A magnet hung at 45° with magnetic meridian makes an angle of 60° with the horizontal. The actual value of the angle of dip is -

(A) $\tan^{-1}\left(\sqrt{\frac{3}{2}}\right)$ (B) $\tan^{-1}(\sqrt{6})$

(C) $\tan^{-1}\left(\sqrt{\frac{2}{3}}\right)$ (D) $\tan^{-1}\left(\sqrt{\frac{1}{2}}\right)$

Open solution

Q71. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

An electron with energy 0.1 keV moves at right angle to the earth's magnetic field of 1×10^{-4} Wbm $^{-2}$. The frequency of revolution of the electron will be :

(Take mass of electron = 9.0×10^{-31} kg)

(A) 1.6×10^5 Hz (B) 5.6×10^5 Hz (C) 2.8×10^6 Hz (D) 1.8×10^6 Hz

Open solution

Q72. JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

At a certain place the angle of dip is 30° and the horizontal component of earth's magnetic field is 0.5 G. The earth's total magnetic field (in G), at that certain place, is :

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{2}$ (C) $\sqrt{3}$ (D) 1

Open solution

Q73. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Choose the correct option

(A) True dip is always equal to apparent dip.

(B) True dip is not mathematically related to apparent dip.

(C) True dip is less than the apparent dip.

(D) True dip is always greater than the apparent dip.

Open solution

Q74. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

At an angle of 30° to the magnetic meridian, the apparent dip is 45°. Find the true dip :

(A) $\tan^{-1}\sqrt{3}$ (B) $\tan^{-1}\frac{1}{\sqrt{3}}$

(C) $\tan^{-1}\frac{2}{\sqrt{3}}$ (D) $\tan^{-1}\frac{\sqrt{3}}{2}$

Open solution

Q75. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

A bar magnet of length 14 cm is placed in the magnetic meridian with its north pole pointing towards the geographic north pole. A neutral point is obtained at a distance of 18 cm from the center of the magnet. If B_H = 0.4 G, the magnetic moment of the magnet is : (1G = 10^{-4} T)

(A) 2.880×10^2 J T^{-1} (B) 2.880 J T^{-1}

(C) 2.880×10^3 J T^{-1} (D) 28.80 J T^{-1}

Open solution

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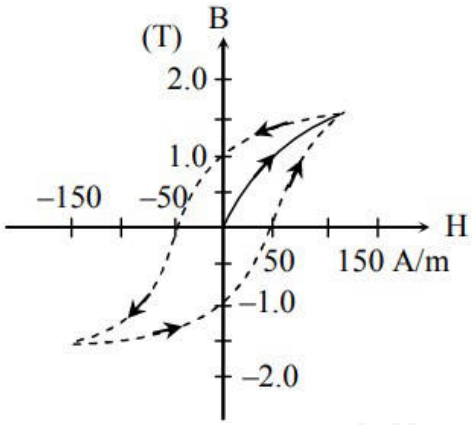
Q76. JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

Magnetic materials used for making permanent magnets (P) and magnets in a transformer (T) have different properties of the following, which property best matches for the type of magnet required?

(A) T : Large retentivity, small coercivity
(B) P : Large retentivity, large coercivity
(C) P : Small retentivity, large coercivity
(D) T : Large retentivity, large coercivity

Open solution

Q77. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)



The figure gives experimentally measured B vs H variation in a ferromagnetic material. The retentivity, coercivity and saturation, respectively, of the material are :

(A) 1.0 T, 50A/m and 1.5 T
(B) 1.5 T, 50 A/m and 1.0 T
(C) 1.5 T, 50 A/m and 2.0 T
(D) 150 A/m, 1.0 T and 1.5 T

Open solution

Q78. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

A magnetic compass needle oscillates 30 times per minute at a place where the dip is 45°, and 40 times per minute where the dip is 30°. If B_1 and B_2 are respectively the total magnetic field due to the earth at the two places, then the ratio $\frac{B_1}{B_2}$ is best given by :

(A) 1.8
(B) 2.2
(C) 0.7
(D) 3.6

Open solution

Q79. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

At some location on earth, the horizontal component of earth's magnetic field is 18×10^{-6} T. At this location, magnetic needle of length 0.12 m and pole strength 1.8 Am is suspended from its mid-point using a thread, it makes 45° angle with horizontal in equilibrium. To keep this needle horizontal, the vertical force that should be applied at one of its ends is -

(A) 3.6×10^{-5} N
(B) 1.8×10^{-5} N
(C) 1.3×10^{-5} N
(D) 6.5×10^{-5} N

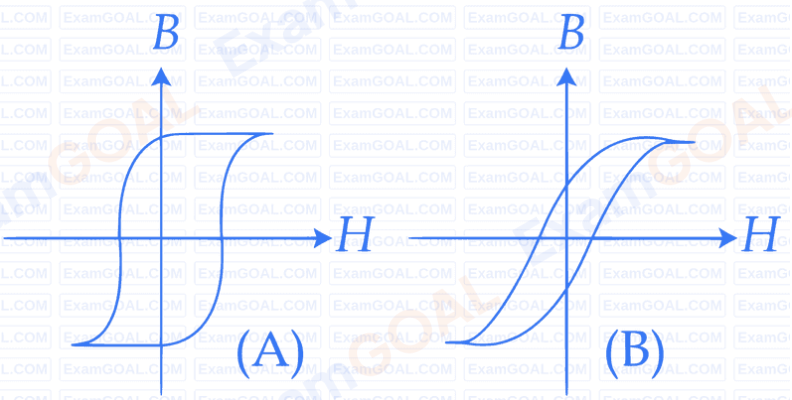
Open solution

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Q80. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

Open solution

Hysteresis loops for two magnetic materials A and B are given below :



These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use :

(A) A for transformers and B for electric generators.

(B) B for electromagnets and transformers.

(C) A for electric generators and transformers.

(D) A for electromagnets and B for electric generators.

Q81. JEE Main 2014 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Open solution

The magnetic field of earth at the equator is approximately 4×10^{-5} T. The radius of earth is 6.4×10^6 m. Then the dipole moment of the earth will be nearly of the order of :

(A) 10^{23} A m^2 (B) 10^{20} A m^2 (C) 10^{16} A m^2 (D) 10^{10} A m^2

Q82. JEE Main 2013 (Online) 25th April Morning Slot · MCQ (Single Correct Answer)

Open solution

The earth's magnetic field lines resemble that of a dipole at the centre of the earth. If the magnetic moment of this dipole is close to 8×10^{22} Am^2, the value of earth's magnetic field near the equator is close to (radius of the earth = 6.4×10^6 m)

(A) 0.6 Gauss (B) 1.2 Gauss (C) 1.8 Gauss (D) 0.32 Gauss

Q83. AIEEE 2003 · MCQ (Single Correct Answer)

Open solution

A thin rectangular magnet suspended freely has a period of oscillation equal to T. Now it is broken into two equal halves (each having half of the original length) and one piece is made to oscillate freely in the same field. If its period of oscillation is T', the ratio $\frac{T'}{T}$ is

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 2 (D) $\frac{1}{4}$

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Q84. JEE Main 2023 (Online) 12th April Morning Shift · **Numerical**

A compass needle oscillates 20 times per minute at a place where the dip is 30° and 30 times per minute where the dip is 60°. The ratio of total magnetic field due to the earth at two places respectively is $\frac{4}{\sqrt{x}}$. The value of x is

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **D**	3. **A**	4. **D**	5. **B**
6. **A**	7. **A**	8. **B**	9. **B**	10. **C**
11. **A**	12. **C**	13. **A**	14. **A**	15. **D**
16. **B**	17. **C**	18. **A**	19. **C**	20. **A**
21. **A**	22. **C**	23. **A**	24. **B**	25. **B**
26. **C**	27. **B**	28. **A**	29. **A**	30. **C**
31. **C**	32. **B**	33. **D**	34. **D**	35. **A**
36. **A**	37. **B**	38. **A**	39. **B**	40. **A**
41. **B**	42. **B**	43. **D**	44. **C**	45. **B**
46. **C**	47. **B**	48. **B**	49. **D**	50. **A**
51. **B**	52. **B**	53. **A**	54. **D**	

Numerical

55. **12**	56. **64**	57. **28**	58. **10**	59. **6**
60. **6**	61. **2**	62. **250**	63. **1**	64. **8**
65. **22**				

MCQ (More than One Correct Answer)

66. **C, D**

Out of syllabus

67. **D**	68. **A**	69. **B**	70. **A**	71. **C**
72. **A**	73. **C**	74. **D**	75. **B**	76. **B**
77. **A**	78. **C**	79. **D**	80. **B**	81. **A**
82. **D**	83. **B**	84. **243**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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